

Abstract

Physical and structural tree measurements are applied in forestry, precision agriculture and conservation for various reasons. Since measuring tree properties manually is tedious, measurements from only a small subset of trees present in a forest, agricultural land or survey site are often used. Utilizing robotics to autonomously estimate physical tree dimensions would speed up the measurement or data collection process and allow for a much larger set of trees to be used in studies. In turn, this would allow studies to make more generalizable inferences about areas with trees. To this end, a system to generate and refine the semantic representation of the topology of trees in real-time has been developed.

Methodology

Six contributions have been made while developing the semantic tree mapping pipeline.

Two contributions were made through the simulation and real-world setup:

1. A simulation environment with a leafless tree and a lidar mounted drone to develop and test the proposed tree mapping pipeline in a simplified scenario was created
2. A hardware sensor rig equipped with state-of-the-art sensors to collect pose estimates and 3D point cloud data for semantic tree mapping in the real world was assembled

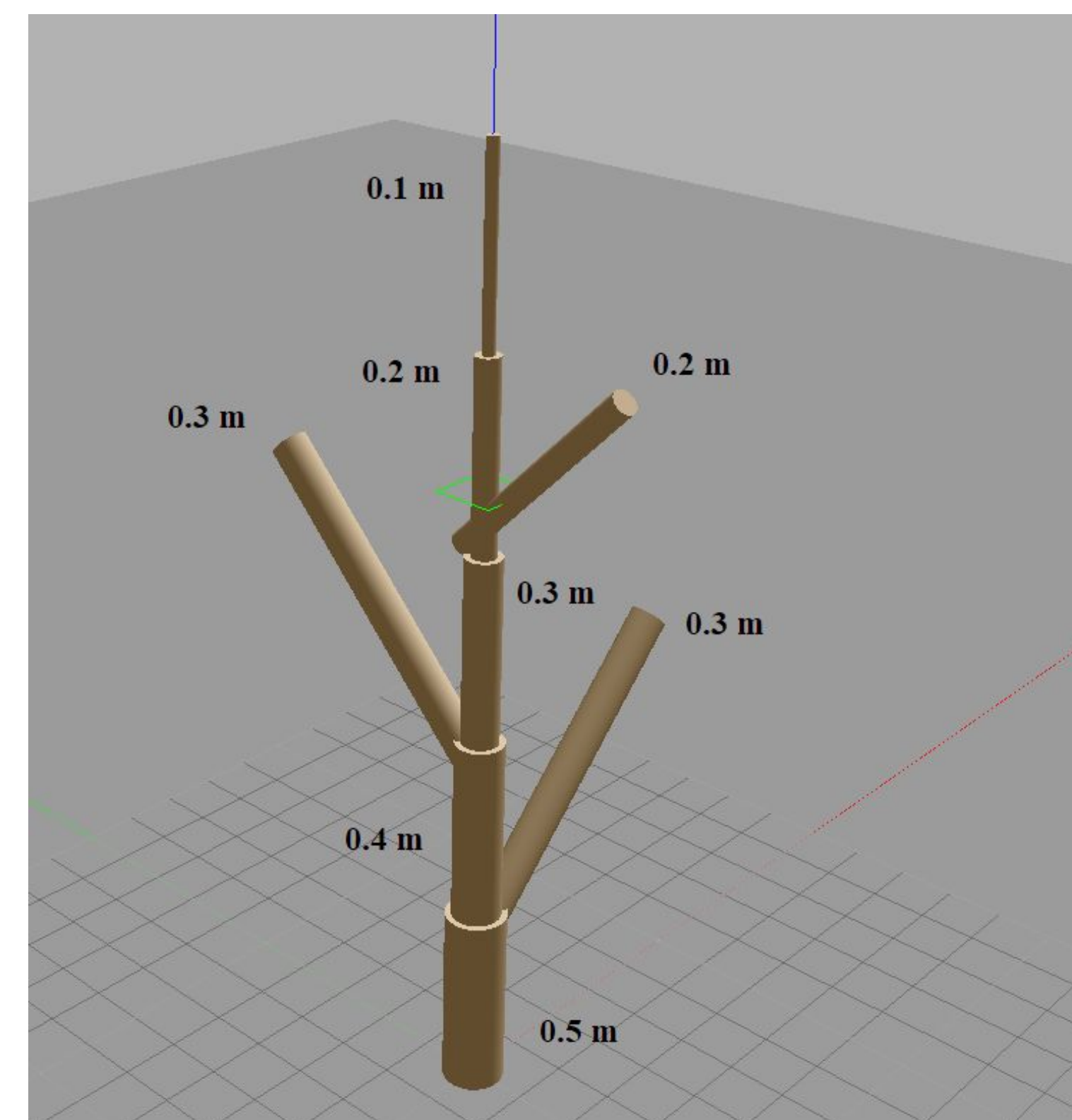
Four contributions were made through the four stages of the core mapping algorithm:

1. A lidar-based deep learning network to identify woody portions of a tree like trunk and branches while ignoring surrounding objects was trained
2. An algorithm to split the identified woody portions of the tree into individual trunk and branch segments was developed
3. An optimization framework to parameterize each trunk and branch segment into uniquely identifiable cylinders that collectively represent a portion of the semantic tree map was implemented
4. A multi-sensor factor graph that incorporates two forms of odometry and incrementally assembles the semantic tree map and optimizes its parameters over time was proposed and utilized

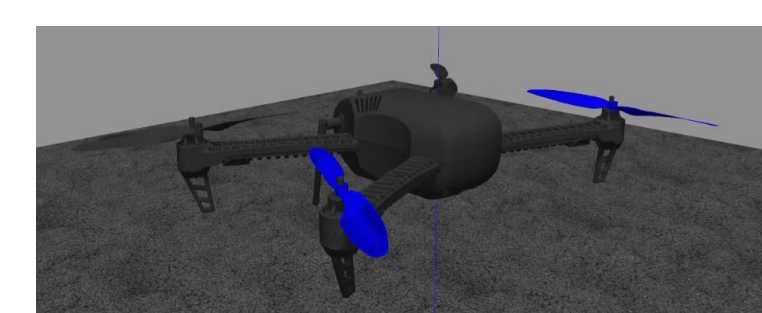
Stages of Mapping Pipeline

1. **Tree Detection**: Semantic segmentation network based on ERF-Net architecture and input layer from RangeNet++ applied on lidar range images to detect non-leafy woody cylindrical parts of a tree
2. **Trunk and Branch Separation**: Custom pair-wise distance thresholding algorithm applied on lidar range images to separate individual trunk and branch instances among detected parts of a tree
3. **Trunk and Branch Parametrization**: Non-linear optimization using C++ Ceres Solver to generate unique 3D cylinder landmark features such that trunk and branch pointcloud lay on the cylinder surface
4. **Map Assembly and Refinement**: Multi-sensor factor graph using custom graph nodes in C++ GTSAM to incrementally assemble and refine the generated tree map in real-time

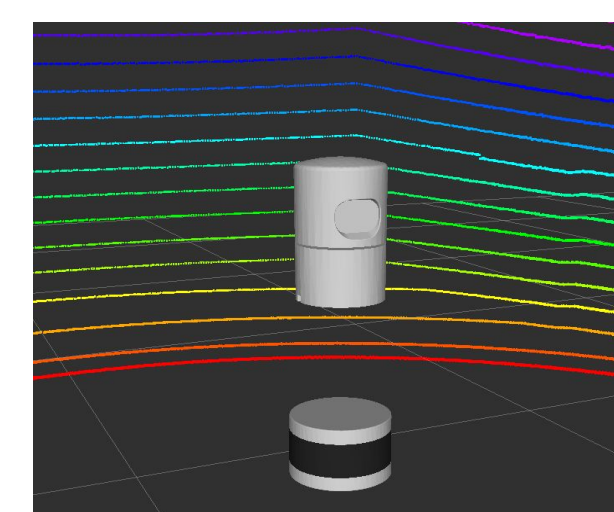
Simulation Setup



Leafless Tree



Iris Drone



Velodyne Lidars

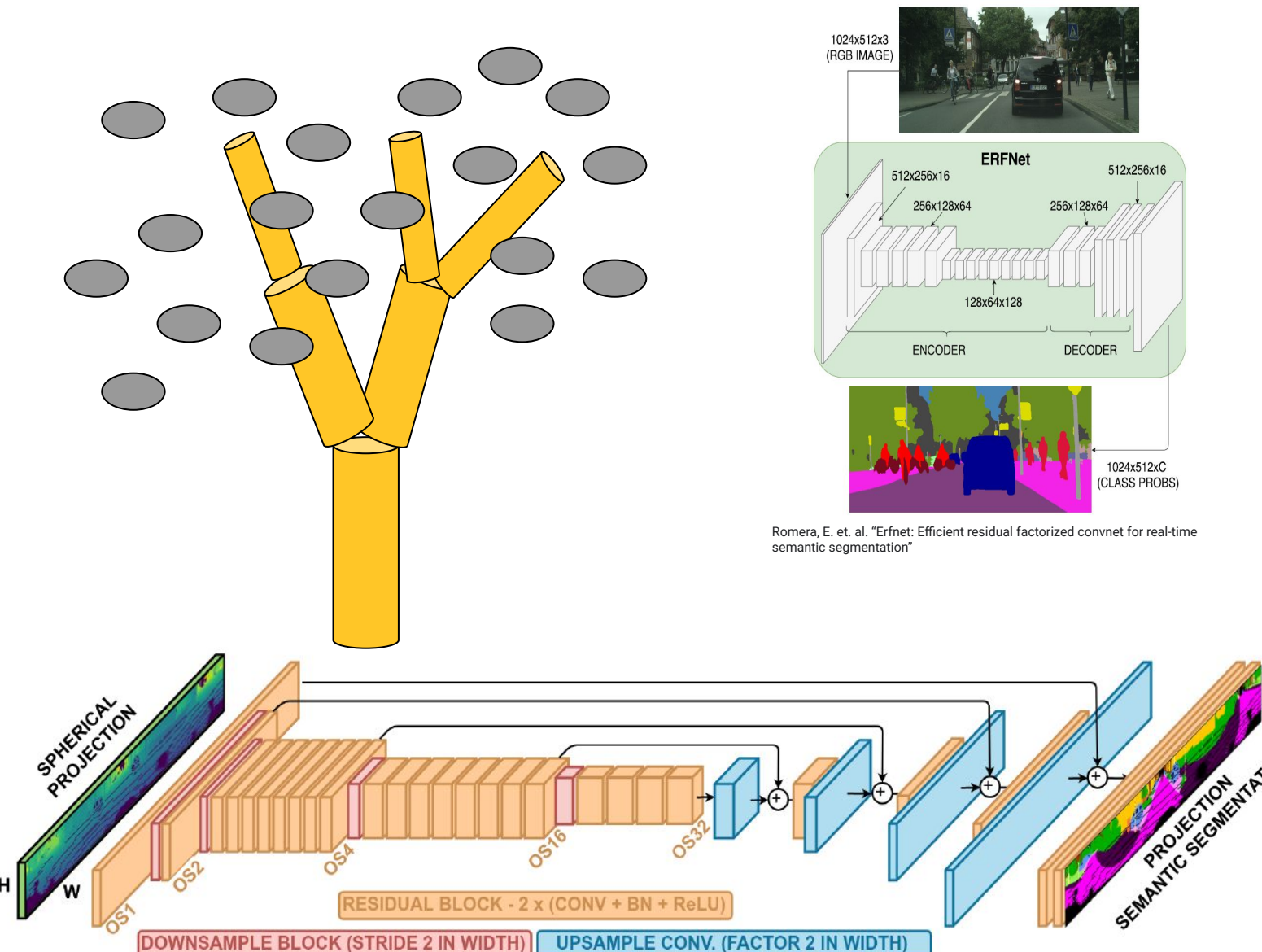
Real-World Setup



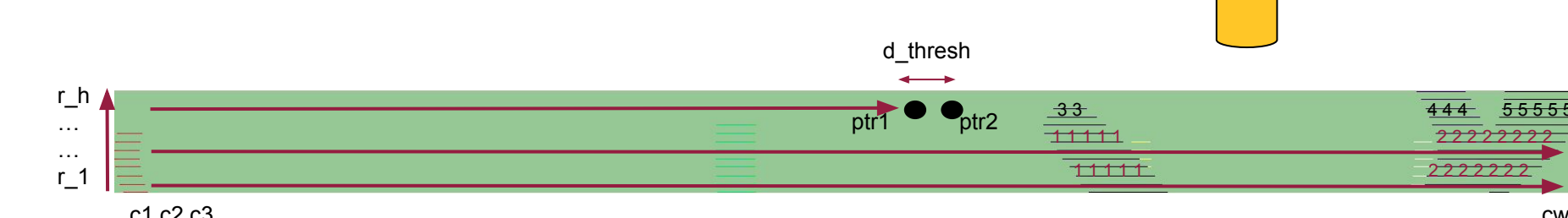
Handheld Sensor Rig

Mapping Pipeline

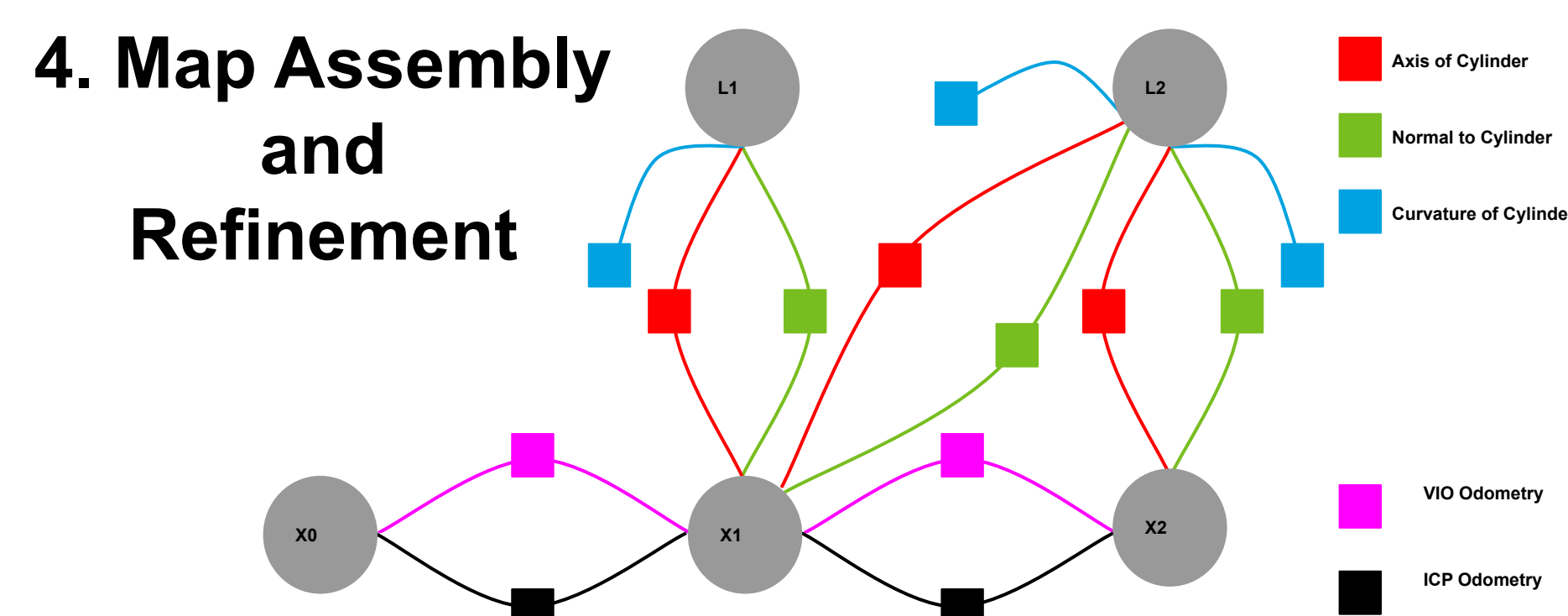
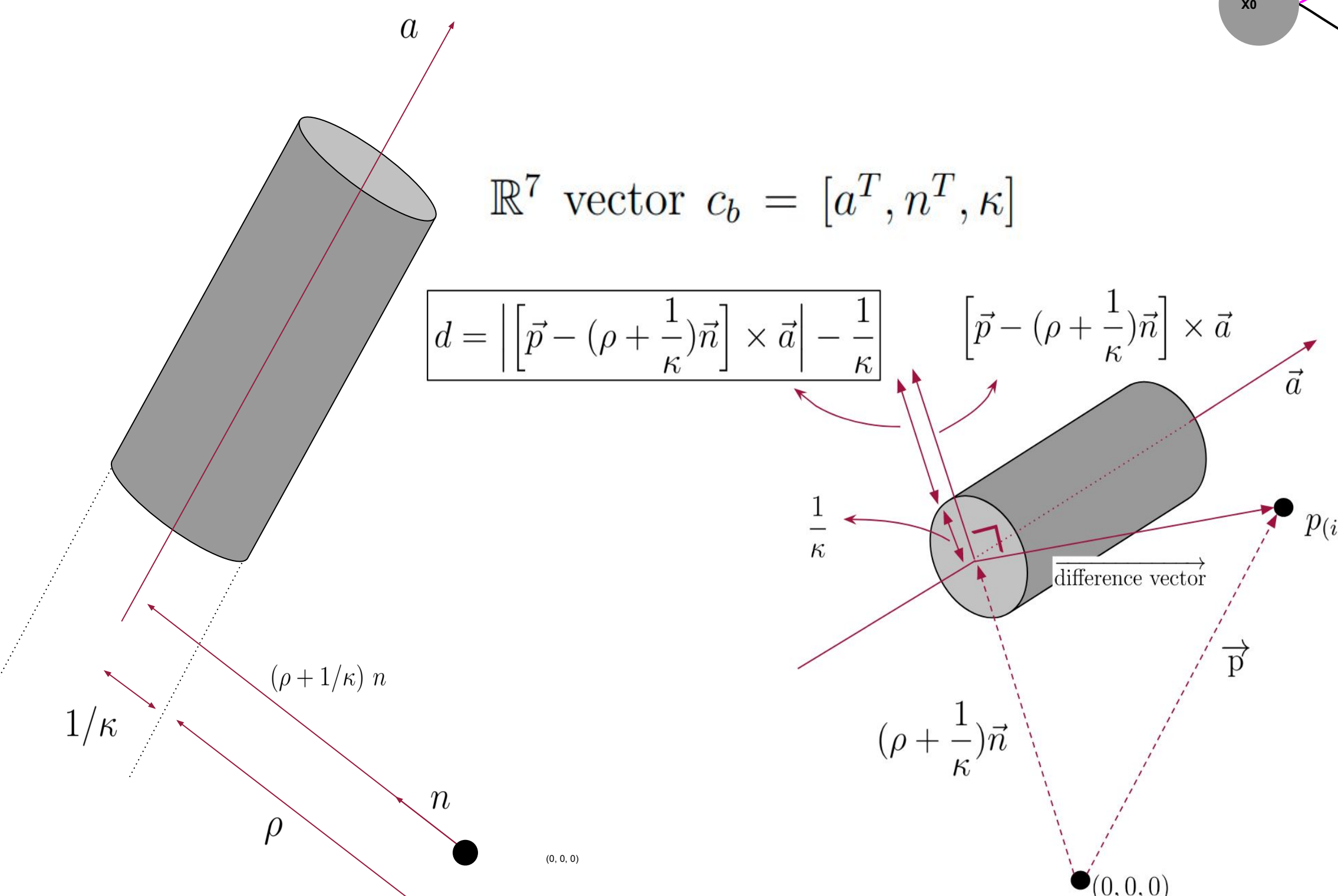
1. Tree Detection



2. Trunk and Branch Separation



3. Trunk and Branch Parameterization



$$e = f(L, X)$$

$$e = T_g^T L_g - L_l$$

$$e = \mathcal{R}_{7 \times 7} (L_g - t_{7 \times 1}) - L_l$$

$$e = \begin{bmatrix} \mathcal{R}_{3 \times 3} & 0_{3 \times 3} & 0 \\ 0_{3 \times 3} & \mathcal{R}_{3 \times 3} & 0 \\ 0_{1 \times 3} & 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} L_g \\ L_l \\ X_{6 \times 1} \end{bmatrix} - L_l$$

$$e_a = T_g^T L_{ga} - L_{la}$$

$$e_a = \mathcal{R}_{3 \times 3} (L_{ga} - 0_{3 \times 1}) - L_{la}$$

$$e_a = \mathcal{R}_{3 \times 3} L_{ga} - L_{la}$$

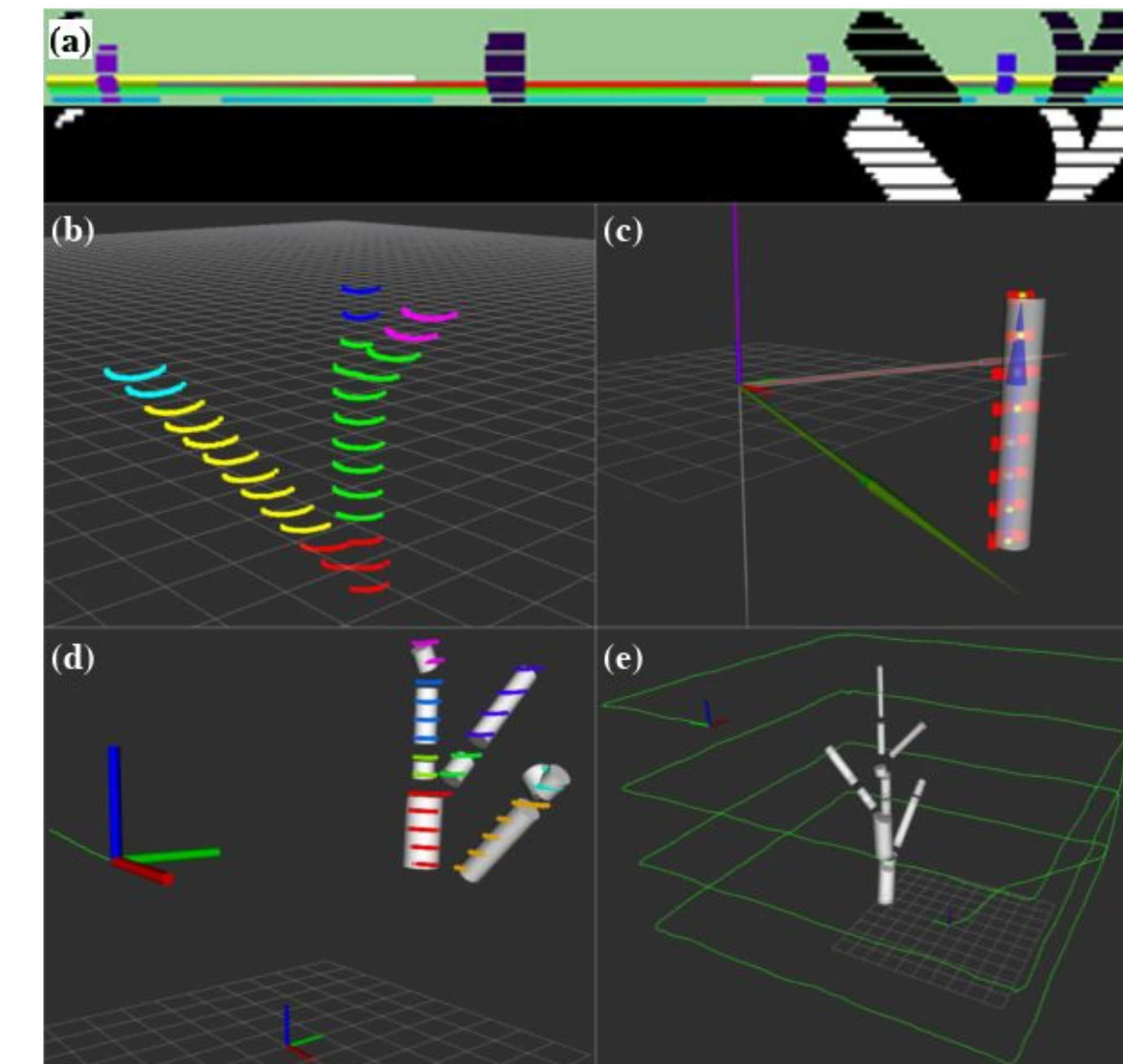
$$e_n = T_g^T L_{gn} - L_{ln}$$

$$e_n = \mathcal{R}_{3 \times 3} (L_{gn} - x_{3 \times 1}) - L_{ln}$$

$$e_k = T_g^T L_{gk} - L_{lk}$$

$$e_k = L_{gk} - L_{lk}$$

Simulation Results



Deep Learning Results:

- Verified on simulation data
- Data Set: 396 lidar scans
- Batch size: 3, Epochs: 7
- Range Image: 32 x 360
- Input: XYZ channels only (intensity was constant)
- Output: Segmented mask
- Encoder trained first.
- Encoder training output layer: 4 x 180 (1/8 scale)
- Decoder trained with frozen encoder
- Training accuracy: 96%

Trunk and Branch Parameterization Results:

- Error metric
 - Distance metric from cylinder fitting equation
 - Radius percentage error corresponding to radius in Gazebo SDF model file of tree
- Simulation
 - Data set: 10 scans, 24 cylinders
 - Maximum Error in Radius: 2% ≈ 0.5 cm

Real-World Results

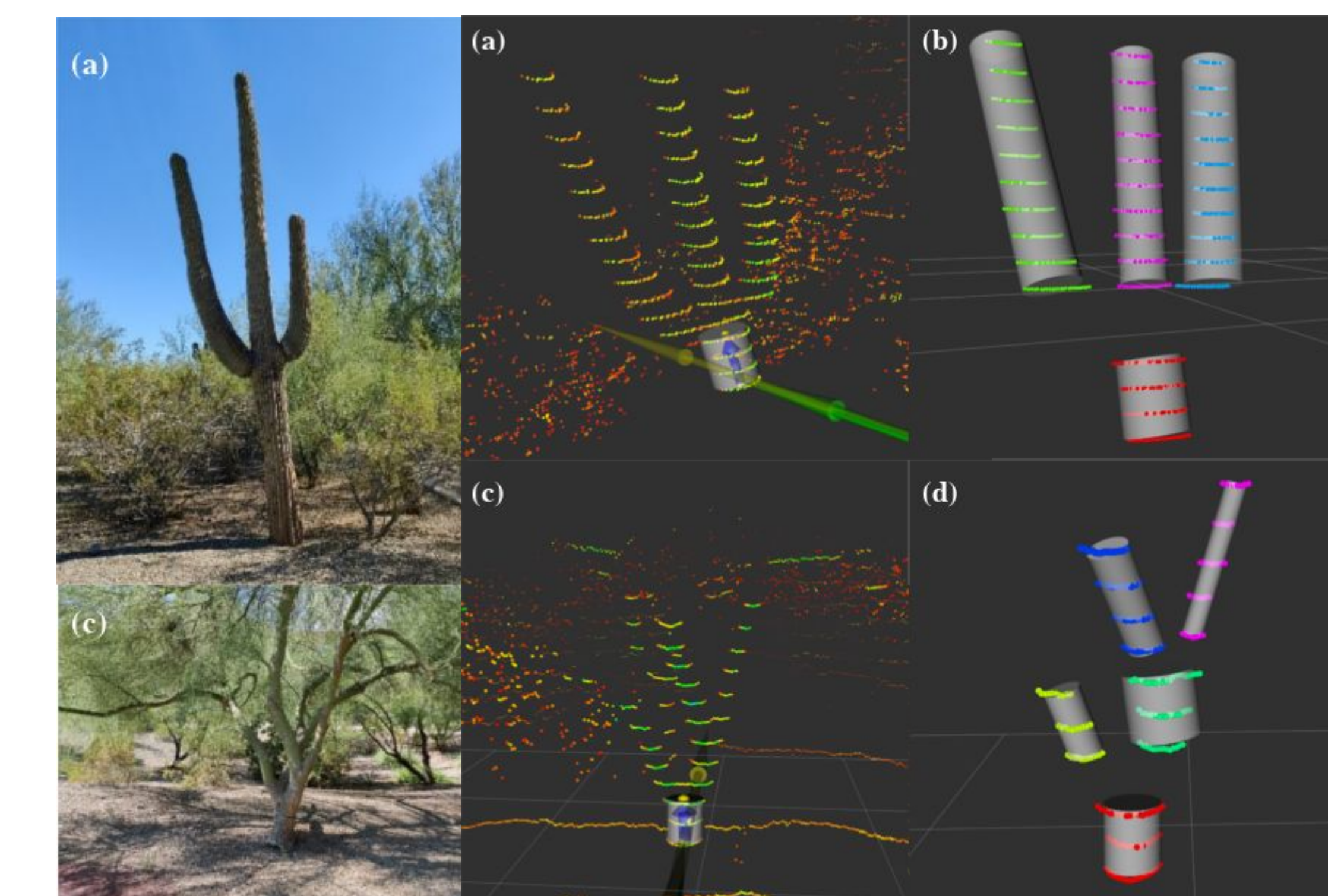


Table 5.1: Radius (cm) of different segments of saguaro.

Segment Name	Tape Measure (cm)	Cylinder Fitting (cm)
Base	23.8 - 25.8	25.99
Left	> 21.5	23.73
Center	> 16.8	19.25
Right	> 16.6	23.91

Table 5.2: Radius (cm) of different segments of palo verde.

Segment Name	Tape Measure (cm)	Cylinder Fitting (cm)
Base	14.95 - 16.17	14.51
Left	7.68 - 8.49	8.79
Right	10.91 - 11.72	12.56
RightLeft	9.5	9.07
RightRight	5.66 - 6.47	5.24

Acknowledgement

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